

Number	Question	Answer
1	What is an active transducer that generates an electrical signal and does not need external power?	Thermocouple
2	What is a rugged, specialized industrial computer used to automate and control electromechanical processes?	Programmable Logic Controller (PLC)
3	What is an electroacoustic transducer which converts mechanical energy to electrical energy?	Piezoelectric Transducer
4	What affects ADC performance?	Number of bits (resolution)
5	What does "IoT Based BMS" refer to?	Data Acquisition and Control System
6	In an ADC, what parameter describes the maximum input frequency to prevent aliasing?	Nyquist Rate / Sampling Rate
7	In an ADC, why does quantization error occur?	Finite resolution (limited number of bits)
8	Which type of ADC converts an analog input into a digital output using a successive approximation method?	Successive-Approximation Register ADC (SAR ADC)
9	What is the simplified form of the Boolean expression $A'C + A'B + AB$?	$A'C + B$ (This simplifies to a 2-input OR gate for certain inputs, but the expression is the direct answer)
10	What is a common PLC programming language for timers?	Ladder Logic or Function Block Diagram (FBD)
11	What is the primary programming language for PLCs?	Ladder Diagram (Ladder Logic)
12	What is the time taken for the PLC to read inputs, execute logic, and update outputs called?	Scan Cycle / Process Cycle Time
13	What is a graphical programming language used to program Programmable Logic Controllers (PLCs)?	Ladder Logic / Ladder Diagram
14	What is a text-based PLC programming language that uses mnemonic codes such as ADD and MOV?	Instruction List (IL)
15	How can you automate the lighting and AV systems in conference rooms of a smart building?	Install an IoT-based BMS controller
16	What are active transducers that do not need external power for temperature measurement?	Thermocouples
17	What are Van der Waals forces?	Weak, attractive intermolecular forces arising from temporary or permanent dipoles.
18	What do ABCD parameters represent in a two-port network?	They are transmission parameters used to model cascaded networks.

19	What is a transducer that converts mechanical force to an electrical signal?	Load Cell
20	What is an electrical device containing junctions of two dissimilar metals, used as a temperature sensor?	Thermocouple
21	In a centralized Building Management System (BMS), which components are integrated for high security?	All of the above (Access Control, Surveillance, Alarms)
22	What is a passive transducer that requires an external power source?	Resistive Strain Gauge
23	What system automatically detects suspicious actions for high security?	AI-Based Video Analytics System
24	What is a common application of DSP involving instant sound processing to improve clarity or quality?	Real-time Audio Enhancement
25	What device converts an analog audio signal to a digital signal?	Analog-to-Digital Converter (ADC)
26	What device takes one audio input and provides multiple outputs with real-time enhancement?	Audio Mixer / Digital Audio Workstation (DAW)
27	What key technology enables seamless communication between HVAC, lighting, and security systems?	IoT-Based Building Management System (IoT-BMS)
28	In an HVAC system, why is a Variable Air Volume (VAV) system preferably used?	To adjust airflow based on demand, improving efficiency.
29	What is the advantage of a Variable Air Volume (VAV) system over a Constant Air Volume (CAV) system?	Lower Energy Consumption
30	What is a control strategy that uses outdoor air for cooling when conditions are favorable?	Economizer Control
31	Why is it essential to integrate a fire alarm system with an access control system in a high-rise building?	For automatic door unlocking, elevator capture, and safe egress.
32	What type of logic gate's output is 1 only when all of its inputs are 1?	AND Gate
33	What does De Morgan's Theorem state?	$(X \cdot Y)' = X' + Y'$ and $(X + Y)' = X' \cdot Y'$
34	What is an indoor device used to detect the presence of a person to automate systems like lighting and HVAC?	Occupancy Sensor
35	What are the advantages of using an IoT-Based BMS over a traditional BMS?	All of the above (Data Analytics, Cloud)

		Connectivity, Remote Monitoring & Control)
36	What is a method for protecting data during transmission using encryption and authentication?	Secure Communication Protocol (e.g., TLS/SSL)
37	Which methods can be used to verify a person's identity for access control?	All of the above (PIN, Face ID, Fingerprints/Biometrics)
38	Which logic gate is a universal gate that can define all other logic gates?	NAND Gate
39	What is a text-based PLC programming language used for complex data manipulation and looping?	Structured Text (ST)
40	In digital electronics, what defines the relationship between inputs and outputs?	Truth Table
41	For three variables (A, B, C), how many combinations are in a truth table?	8
42	How many cells are there in a 3-variable K-map?	8
43	In a 4-variable K-map, how many cells are available for simplification?	16
44	What is the advantage of a K-Map over a Truth Table for Boolean simplification?	It provides a visual method to minimize expressions.
45	What is a group of adjacent 1s in a K-map called?	An Implicant (or specifically, a Prime Implicant)
46	Which logic gate outputs 1 only when all of its inputs are 0?	NOR Gate
47	In an ASM chart, what symbol indicates the direction of state transition?	An Arrow
48	What law in Boolean algebra is represented by $A + A = A$ and $A \cdot A = A$?	Idempotent Law
49	When the J and K inputs of a JK flip-flop are tied together, what does it function as?	T (Toggle) Flip-Flop
50	What is the distributive law in Boolean algebra?	$A(B + C) = AB + AC$
51	What is an example of the complementary law in Boolean algebra?	$A + A' = 1$
52	When will a 3-input AND gate output a 1?	When all three inputs are 1.
53	What kind of flip-flop is used to make a divide-by-two counter?	T (Toggle) Flip-Flop
54	How many cells are there in a 4-variable K-Map?	16
55	A mod-4 binary counter has how many states?	4
56	In a D flip-flop, when does the output Q follow the input D?	On the active clock edge (or when the clock triggers).
57	Which is a valid absorption law in Boolean algebra?	$A + A \cdot B = A$
58	How many flip-flops are needed for a 5-bit binary counter?	5

59	Which type of flip-flop has an output that changes state on the clock edge based on the D input?	D Flip-Flop
60	In a flowchart, what symbol represents a point where a condition is tested?	Decision Box
61	In an ASM chart, what does a state box represent?	The current state and its unconditional outputs.
62	Which logic gate outputs 1 when all inputs are equal?	XNOR Gate
63	In an ASM chart, what produces outputs only when a specific condition is true?	Conditional Output Box
64	In ASM design, which diagram represents outputs that depend only on the current state?	Moore Machine
65	In ASM design, why is the clock signal important in a synchronous process?	To synchronize state transitions and prevent hazards.
66	What is the design of an Arithmetic Logic Unit (ALU) based on?	Combinational Logic
67	In asynchronous circuits, what is a momentary, unwanted output change called?	Hazard / Glitch
68	What type of state machine depends on both the present state and current input for its output?	Mealy Machine
69	What type of register in a microprocessor points to the top of the stack?	Stack Pointer
70	Which type of memory uses transistors to store data without needing refreshing?	SRAM (Static Random-Access Memory)
71	What type of data hazard occurs when an instruction reads an operand before it is updated by a previous instruction?	Read-After-Write (RAW) Hazard / Data Dependency
72	What is the process where the CPU saves the state of a process and loads another?	Context Switching
73	In microarchitecture, which is typically NOT a programmer-visible register?	Memory Data Register (MDR) / Internal Buffer Register
74	What is the ratio of output to input in the Laplace domain called?	Transfer Function
75	What bus transfers data to and from the ALU?	Data Bus
76	What register stores the current instruction being executed?	Instruction Register
77	What is the output of a 3-input OR gate if all inputs are 0?	0
78	What type of RAM uses one transistor and one capacitor per memory cell?	DRAM (Dynamic RAM)

79	How many bits does an 8-byte addressable register have?	64 bits
80	Which logic gate outputs 1 when at least one of its inputs is 1?	OR Gate
81	What type of register is used for temporary storage and arithmetic operations?	General-Purpose Register
82	What software instructs the computer what to do when it boots up?	BIOS (Basic Input/Output System)
83	What CPU register stores the address of the next instruction?	Program Counter (PC)
84	How many bits can a 64-bit register hold?	64 bits
85	Which statement is NOT true about Moore's Law?	It states that transistor current increases with frequency.
86	Which memory type is denser and uses less chip area per bit?	DRAM
87	Which data hazard occurs when an instruction tries to write to a location before a previous instruction reads from it?	Write-After-Read (WAR) Hazard
88	Which flag indicates an arithmetic carry or borrow from the most significant bit?	Carry Flag
89	What architecture uses registers for operations without going through memory?	Load/Store Architecture (RISC)
90	Which hazard occurs when two instructions try to write to the same location out of order?	Write-After-Write (WAW) Hazard
91	What is the maximum unsigned integer a 16-bit register can hold?	65,535
92	What register holds the address of the memory location to be accessed?	Memory Address Register (MAR)
93	What hazard occurs when the outcome of a branch instruction is not yet known?	Control Hazard (Branch Hazard)
94	How many input channels does a multiplexer with 5 select lines support?	32
95	What is an open-source platform for building electronics projects?	Arduino
96	Which is NOT true for a Load/Store (R2R) architecture?	Memory is accessed before every arithmetic operation.
97	What architecture uses a single memory space for instructions and data?	Von Neumann Architecture
98	What is a communication method that uses encryption to protect data during transmission?	Secure Communication Protocol (e.g., TLS/SSL, SSH)
99	In a T flip-flop, when does the output change state?	When the T input is HIGH and the clock triggers.

100	In asynchronous circuits, what is an unpredictable state due to variable delays called?	Race Condition
101	In a JK flip-flop, what happens when J=1 and K=1?	The output toggles (changes state) on the clock edge.
102	What is a logical function that can be used to detect an odd number of 1s in the input?	XOR Gate
103	What flip-flop stores binary data by capturing the input on a clock pulse?	D Flip-Flop
104	What is a state table for a controller?	A list of present states, inputs, and their corresponding next states and outputs.
105	In an ASM chart, what does the first level of design with a multiplexer determine?	The next state for the registers.
106	In an ASM chart, what is the box that tells the effect of an input on the control subsystem called?	Decision Box
107	Which of the following is not a type of register?	Latch (A latch is a level-triggered storage element, often distinguished from an edge-triggered register/flip-flop)
108	What is the advantage of using R2R architecture?	It reduces complexity by minimizing memory access for operations.
109	What is the purpose of the Arduino delay() function?	To pause the program for a specified amount of time (in milliseconds).
110	What is the advantage of using a block diagram in control systems?	To provide a clear, visual representation of complex systems, simplifying analysis and design.
111	What is Moore's Law?	The prediction that the number of transistors on a microchip doubles approximately every two years.
112	What is a disadvantage of Harvard Computer Architecture?	Increased hardware complexity due to separate instruction and data memories/buses.
113	How many flip-flops are required for a 5-bit counter?	5

114	In the transfer function $F(s) = 5/(s^2 + s + 6)$, what is the order of the system?	Second Order System
115	In the transfer function $F(s) = 5/(s + 2)$, what is the order of the system?	First Order System
116	What does a signal flow graph represent?	System variables and their interdependence.
117	What does the Arduino command <code>digitalWrite(pin, HIGH);</code> do?	Sets the specified digital pin to a high voltage level (ON).
118	How many inputs can a multiplexer with 3 select lines support?	8 inputs
119	In optical communication, why is a PIN diode preferred?	Its wide intrinsic layer allows for a faster response time and higher bandwidth.
120	In hybrid (h) parameters, what does h_{fe} represent?	Forward current gain in a common-emitter configuration (β).
121	In an electromagnetic wave in free space, what is the relationship between the E and B fields?	They are perpendicular to each other and to the direction of propagation.
122	What is the purpose of Delta-Star (Δ -Y) and Star-Delta (Y- Δ) transformations?	To simplify complex resistor networks for analysis.
123	What is a Wheatstone Bridge used for?	To measure an unknown electrical resistance.
124	In a resistive network, the efficiency of power transfer is 50%. What condition must be true?	The load resistance is equal to the source resistance (Impedance Matching).
125	In a second-order system, what condition causes oscillations before reaching steady state?	When the damping ratio is less than 1 (underdamped).
126	When does dielectric breakdown occur?	When the electric field exceeds the dielectric strength of the material.
127	What is the nature of magnetic susceptibility in diamagnetic materials?	It is negative and very small.
128	At what temperature does a ferromagnetic material lose its ferromagnetic properties?	Above its Curie temperature.
129	What happens to the capacitance when a dielectric is placed between two metal plates?	It increases.
130	What property should a dielectric material have for a high-voltage capacitor in an electric vehicle?	High dielectric strength and high dielectric constant.
131	Which material is used for a transformer core to minimize hysteresis and eddy current losses?	Soft iron with silicon alloying (Silicon Steel).
132	When system blocks are connected in cascade, what is the overall transfer function?	The product of the individual transfer functions.

133	What is the instruction set for a microprocessor that is converted into binary called?	Machine Language / Machine Code
134	In a JK flip-flop, what is the output when J=0 and K=1?	The output resets to 0.
135	What is the primary purpose of Boolean algebra in logic circuits?	To simplify and analyze logic expressions and circuits.
136	Which gates are considered universal gates?	NAND and NOR Gates
137	What are the only two variable values used in Boolean functions?	0 and 1
138	How many designated bits (cells) are there in a 4-variable K-map?	16
139	In a D flip-flop, the output Q follows the input D when what occurs?	On the active clock edge.
140	How many flip-flops are required to design a 5-bit binary counter?	5 Flip-flops
141	Which type of flip-flop has an output that changes state depending on its previous inputs?	JK Flip-Flop (or T Flip-Flop). D Flip-Flop depends on the current input.
142	What is the process of storing and restoring process information so CPU execution can resume later?	Context Switching
143	Which theorem states that the response in a linear circuit is the sum of responses from each independent source?	Superposition Theorem
144	At resonance in a series RLC circuit, what is the total impedance?	Minimum and equal to the resistance (R).
145	In a parallel RLC circuit at resonance, what is the voltage across the RLC combination?	Maximum.
146	In a series RLC circuit, when does resonance occur?	When inductive reactance (X_L) equals capacitive reactance (X_C).
147	In Z-parameters for a two-port network, what does Z_{21} represent?	The transfer impedance (open-circuit).
148	Thevenin's theorem states that any linear bilateral network can be replaced by what?	A single voltage source in series with a single resistance.
149	When connecting two amplifiers in cascade, what should the input impedance of the second amplifier be?	High, to avoid loading the first amplifier.
150	Which theorem states that the current in a branch due to a voltage source is the same if their positions are interchanged?	Reciprocity Theorem
151	In a PUT RC timing circuit, what happens if the capacitor value is increased?	The time delay before the PUT triggers increases.

152	Which law states that a time-varying magnetic field induces an electric field?	Faraday's Law of Electromagnetic Induction
153	What is the relationship between real power (P), reactive power (Q), and apparent power (S)?	$S^2 = P^2 + Q^2$
154	Which law states that the curl of the electric field is equal to the negative rate of change of the magnetic field?	Faraday's Law of Electromagnetic Induction (in differential form)
155	How can you improve the power factor of an inductive AC circuit?	By adding a parallel capacitor.
156	Coulomb's law states that the electrostatic force is directly proportional to what?	The product of the magnitudes of the two charges.
157	In a common-source FET amplifier, what is the effect of increasing the source (degeneration) resistor?	It reduces voltage gain but improves linearity and stability.
158	According to Maxwell's first equation (Gauss's law for electricity), what is the electric flux related to?	The net electric charge enclosed within the surface.
159	If a conducting loop is placed in a steady (DC) uniform magnetic field, what is the induced EMF?	Zero
160	A cylindrical conductor carries a steady current with a non-uniform current density $J(R)$ that increases with radius R . What is true about the magnetic field inside?	The magnetic field at a radius R is proportional to the current enclosed within that radius.
161	In a full-wave rectifier with a center-tapped transformer, if the transformer winding resistance is high, what happens to the DC output voltage?	It becomes lower than the theoretical value due to increased voltage drop.
162	Why do MOSFETs have a higher input impedance than BJTs?	Because the MOSFET gate is insulated (capacitively coupled), drawing negligible DC current.
163	In a full-wave rectifier with a filter capacitor, the DC output voltage drops significantly. What is a likely cause?	The filter capacitor value is too small, causing excessive ripple.
164	In a Zener regulator under load, the output voltage is higher than desired. What is a likely cause?	The Zener diode is not operating in the breakdown region (or it is open).
165	In a half-wave rectifier, the output voltage is nearly zero. What is the most probable cause?	The diode is open-circuited.
166	In a voltage doubler circuit, the output is significantly lower than theoretical. What is a reason?	The capacitors are discharging too quickly under load (capacitor value too small or load too heavy).

167	In an N-channel enhancement MOSFET, what primarily controls the drain current (I_D) in the linear (ohmic) region?	The gate-to-source voltage (V_{GS}).
168	What is the advantage of a Schottky diode in a high-frequency rectifier circuit?	It has a lower forward voltage drop and faster switching speed (reduced reverse recovery time).
169	In a peak detector circuit, why is the output voltage lower than the peak input voltage?	Due to the forward voltage drop across the diode.
170	In a voltage multiplier, the measured output is lower than expected. What is a probable cause?	The capacitor values are too small, or the load is drawing too much current.
171	What is the output of a logic gate that is 1 only if both inputs are 0?	NOR Gate
172	What is a flowchart symbol used to represent a process or action?	Process Box / Rectangle
173	What is the function of the <code>delay()</code> instruction in Arduino?	To pause the program for a specified time in milliseconds.
174	What is the primary purpose of an Algorithmic State Machine (ASM) chart?	To design and specify the behavior of sequential logic circuits.
175	What is a logical function that outputs 1 when an even number of inputs are 1?	XNOR Gate (for two inputs). For multiple inputs, it is an even parity generator.
176	What is the main purpose of a Secure Access Protocol?	To protect data during transmission using encryption, authentication, and integrity checks.